Rubrics as Instruments and Optimization Signals

Geist Atlas · Module 8 · Apparatus as Method

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[**METHODS**] Methods / apparatus contribution · **Family:** Apparatus as Method
Claim: Rubric evolution is construct development; the rubric doubles as an optimisation signal for automated prompt tuning, with dimension-targeted gains that partly transfer across models.

Treats the four rubric iterations, the measurement paradox, scoring, and automated prompt tuning as instrument design rather than incidental tooling.

Derived view of `paper-full-2.0.md` v3.0.118 — §5.2, §5.6, §7.7, §7.8. The paper is canonical; edit it there, not here.

5.2 Evaluation Rubric Design (v2.2)

5.2.1 Rubric Derivation: From Ad Hoc to Literature-Informed The v1.0 rubric (14 tutor dimensions, 6 learner dimensions) was constructed through iterative prompting of Claude during Paper 1.0's pilot phase. The process was pragmatic rather than principled: dimensions were proposed by the LLM based on general pedagogical reasoning, refined through trial scoring, and retained if they appeared to discriminate between conditions. No systematic literature review grounded the dimension selection, and the resulting rubric reflected the LLM's implicit model of "good tutoring" rather than validated constructs from learning sciences or educational measurement.

Paper 2.0 replaced this approach with a structured derivation process. A literature review of eight validated evaluation frameworks—MRBench (Maurya et al., 2025), GuideEval (Liu et

*This sentence is the only one actually authored by Liam Magee in this paper.

al., 2025), MathTutorBench (Macina et al., 2025), ICAP (Chi & Wylie, 2014), TRU (Schoenfeld, 2018), the Danielson Framework (Danielson, 2022), Talk Moves (Michaels & O'Connor, 2015), and a cognitive scaffolding rubric (Figueiredo et al., 2025)—plus five dialogue-level frameworks (Alexander's dialogic teaching principles (Alexander, 2018), Nystrand's process indicators (Nystrand, 1997), T-SEDA (Hennessy et al., 2016), LLM-Rubric (Hashemi et al., 2024), and the BEA 2025 shared task) produced a cross-framework dimension mapping identifying which constructs were well-covered (3+ independent frameworks), partially covered (1–2), or unique to our rubric.

Three findings drove the redesign:

1. **Structural conflation.** The v1.0 rubric conflated what the tutor *perceives* (learner state detection), what it *does* (pedagogical strategy), and what it *elicits* (learner reasoning). GuideEval's Perception→Orchestration→Elicitation (P→O→E) decomposition (Liu et al., 2025) provided a principled separation, validated on 5,177 samples with 89–97% human-LLM agreement.

2. **Empirical redundancy.** A dimension-by-dimension audit predicted high correlations among clusters of v1.0 dimensions (e.g., `relevance` + `personalization` + `memory_integration`; `productive_struggle` + `transformative_potential`). PCA on 1,584 per-turn observations supported this: PC1 explained 80.7% of variance across the 14 dimensions (KMO = 0.938), with a mean inter-dimension correlation of $r = 0.776$. The rubric measured fewer independent constructs than its 14 dimensions implied.

3. **Missing constructs.** Every reviewed framework except our v1.0 rubric measured *content accuracy* (factual correctness). Additionally, `learner_growth` on the tutor rubric was architecturally unscorable—the judge could not see the learner's next turn.

The redesign consolidated 14 tutor dimensions to 8 using the P→O→E decomposition, added `content_accuracy`, and removed `learner_growth`.

The learner rubric required a parallel restructuring. The v1.0 learner rubric (7 dimensions) suffered from three problems identified in the dimension audit: overlap among `question_quality`, `conceptual_engagement`, and `revision_signals` (all measuring "depth of intellectual engagement" from slightly different angles); a `persona_consistency` dimension (5% weight) that created a structural tension with `revision_signals` (showing genuine growth means departing from a frustrated or confused persona); and a `deliberation_depth` dimension that was only scorable for multi-agent learners, creating an architecture-dependent gap in the scoring instrument. The ICAP framework (Interactive→Constructive→Active→Passive; (Chi & Wylie, 2014)) provided a validated hierarchy for collapsing the overlapping engagement dimensions into a single `engagement_quality` construct with empirically grounded scoring anchors: level 1 (Passive) = paraphrases tutor, confirms understanding; level 3 (Active/Constructive) = generates own interpretations, makes connections; level 5 (Interactive) = co-constructs understanding, challenges tutor's framing. `persona_consistency` was removed (its core concern—does the learner feel real?—is captured by `learner_authenticity`), and `deliberation_depth` was moved to the dedicated deliberation rubric (Section 5.2.4). The resulting 5-dimension learner rubric maps to two validated frameworks: ICAP for engagement levels and T-SEDA's behavioral coding clusters (Hennessy et al., 2016) for revision signals (B-cluster: Build on ideas) and conceptual progression (R-cluster: Make reasoning explicit).

The tutor holistic rubric was radically simplified from 6 to 3 dimensions after consistency analysis showed $r = 0.907$ between per-turn and holistic scores, indicating that most holistic signal was redundant with aggregated per-turn scoring. The three retained dimensions—pedagogical arc, adaptive trajectory, and pedagogical closure—capture exclusively arc-level properties that cannot be assessed from individual turns. Scale criteria across all instruments were redesigned following Yamauchi et al. (Yamauchi et al., 2025): only levels 1, 3, and 5 carry full descriptions, as intermediate-level descriptions have limited impact on LLM judges.

5.2.2 Tutor Per-Turn Rubric The evaluation rubric underwent four iterations (v1.0→v2.0→v2.1→v2.2), each responding to empirical anomalies discovered during analysis (documented in Appendix E). The current v2.2 rubric consolidates 14 dimensions to 8, guided by the GuideEval P→O→E decomposition framework and the study's own empirical dimension clustering. The eight dimensions are:

Category	Dimension	Weight	Mechanism Relevance
Perception (P)	`perception_quality`	15%	Calibration: tutor perceives learner's specific state
Perception (P)	`content_accuracy`	10%	Error correction: factual/domain accuracy
Orchestration (O)	`pedagogical_craft`	15%	Calibration: response construction quality
Orchestration (O)	`elicitation_quality`	15%	Adaptive responsiveness: probing learner reasoning
Orchestration (O)	`adaptive_responsiveness`	10%	Adaptive responsiveness: turn-over-turn modulation
Execution (E)	`productive_difficulty`	10%	Calibration + adaptation: challenge calibration
Execution (E)	`epistemic_integrity`	10%	Error correction: intellectual honesty
Recognition	`recognition_quality`	15%	Calibration + error correction: intersubjective stance

The consolidation from 14→8 dimensions was validated through synthetic calibration (r=0.996 against v2.1 scoring on identical responses), confirming that the reduced rubric preserves discriminability while eliminating ceiling-prone and redundant dimensions.

5.2.3 Learner Per-Turn Rubric The learner rubric mirrors the tutor rubric symmetrically (Section 4, Design Principle), scoring learner responses on five ICAP-anchored dimensions:

Dimension	Weight	What It Measures
`engagement_quality`	25%	Active vs. passive participation

Dimension	Weight	What It Measures
`conceptual_progression`	25%	Depth of conceptual engagement over turns
`revision_signals`	20%	Evidence of position revision in response to tutor
`metacognitive_awareness`	15%	Self-monitoring of understanding
`learner_authenticity`	15%	Persona-consistent, non-formulaic responses

5.2.4 Holistic and Deliberation Rubrics Two additional rubrics assess trajectory-level and process-level quality:

- **Tutor holistic** (3 dimensions: `pedagogical_arc`, `adaptive_trajectory`, `pedagogical_closure`) — scores the full multi-turn dialogue as an arc, capturing qualities invisible to per-turn scoring.
- **Deliberation quality** (6 dimensions, applied symmetrically to tutor and learner ego-superego traces) — scores the quality of internal deliberation for multi-agent cells.

5.2.5 Public-Only Output Scoring (v2.1 Fix) A critical methodological decision: per-turn and holistic judges see ONLY public messages (the delivered tutor response and the learner's external message). Internal ego-superego deliberation is scored separately by the deliberation rubric. This prevents a confound where multi-agent cells receive higher scores simply because the judge sees richer internal reasoning.

5.2.6 Rubric Version Tracking Each scored row records which rubric version was used (`tutor_rubric_version`, `learner_rubric_version`, `dialogue_rubric_version`, `deliberation_rubric_version`), auto-resolved from YAML `version:` fields at write time. This prevents cross-version contamination: v1.0 scores (8,987 backfilled rows) are never mixed with v2.2 scores in the same analysis.

5.6 The Measurement Paradox as Methodology

The measurement paradox — where authentic pedagogical engagement produces lower scores under naive evaluation — is not a limitation but a *methodological finding* about what LLM-as-Judge evaluation measures.

The Paradox When the evaluation judge receives only the tutor's response without dialogue context, it interprets careful scaffolding of authentic confusion as failure. Adding dialogue context to the judge resolves this, confirming the paradox is a measurement artifact rather than a quality decline.

What This Reveals LLM judges optimize for *surface resolution* — visible agreement, smooth interaction, confident answers. Productive struggle looks like failure to a judge that cannot see the learner's internal development. This has implications beyond this study for any LLM-as-Judge evaluation of educational interactions.

Rubric Iteration as Evidence The rubric evolution itself constitutes evidence of construct refinement. Each version responded to specific empirical anomalies:

Version	Anomaly Addressed	Change
v1.0	Baseline design	14 tutor dimensions, standard weights
v2.0	Truncation hallucination (31% of Haiku reasonings)	Anti-truncation instruction; modulation dims reweighted
v2.1	Architecture confound (internal deliberation visible to judge)	Public-only output scoring; separate deliberation rubric
v2.2	Dimension redundancy (ceiling on tone, overlap in struggle dims)	14→8 consolidation guided by GuideEval P→O→E and empirical clustering

This iteration parallels the ego-superego dynamic: initial rubric (ego draft) → anomaly detection (superego critique) → revised rubric (substantive revision, not cosmetic).

7.7 The Rubric as Optimisation Signal: Automated Prompt Tuning

The per-dimension rubric enables a capability that holistic scoring cannot: automated prompt optimisation with interpretable gradient signal. We tested this using a simple hill-climbing loop—an LLM recommender analyzes per-dimension weaknesses from scored evaluation rows, proposes targeted prompt edits, benchmarks the revised prompt, and keeps or reverts based on score—applied to a single frustration scenario (Mood: Frustration to Breakthrough) across both base and recognition prompts.

The experiment forms a $2 \times 2 \times 2$ design: prompt condition (base vs. recognition) $\times$ optimisation (unoptimised vs. autotuned) $\times$ generation model (DeepSeek V3.2 vs. Qwen 3.5 9B). All conditions were judged by Sonnet 4.6. The Qwen 3.5 runs were conducted entirely on a local machine via LM Studio, demonstrating that the optimisation loop is feasible without cloud inference for generation.

Autotuning × Recognition × Model (Sonnet judge, frustration scenario)

Model	Base	Base autotuned	Recognition	Recog. autotuned
DeepSeek V3.2	36.6 (n=9)	41.6 (n=5)	55.2 (n=5)	72.9 (n=5)
Qwen 3.5 9B	35.7 (n=5)	45.0 (n=4)	47.9 (n=5)	71.2 (n=5)

Four findings emerge from the complete matrix.

First, **recognition prompts are more improvable than base prompts, not less**. Autotuning gains on recognition (+17.7 DeepSeek, +23.3 Qwen) substantially exceed gains on base (+5.0 DeepSeek, +9.3 Qwen). Recognition does not saturate the rubric; it provides richer starting material that the optimiser can refine further.

Second, **the effects are super-additive**. On DeepSeek, recognition alone adds +18.6 and base autotuning adds +5.0, but recognition + autotuning adds +36.3—exceeding the sum of parts (+23.6) by 54%. The same pattern holds on Qwen (+35.5 combined vs +21.5 sum). Recognition and prompt engineering operate on complementary dimensions.

Third, **the gap between conditions widens under optimisation**. Unoptimised, recognition leads base by 18.6 points (DeepSeek) and 12.2 points (Qwen). After autotuning, the gap grows to 31.3 and 26.0 respectively. If recognition were reducible to "better prompt writing," automated optimisation should converge the conditions; instead, it diverges them.

Fourth, **recognition autotuning converges across models**. Despite starting from different baselines (DeepSeek 55.2, Qwen 47.9), both models reach a similar autotuned ceiling (~71–73). This model-independent ceiling suggests a structural quality boundary in the recognition prompt that autotuning can unlock regardless of generation capability. The base prompt has a lower ceiling (~41–45) that no amount of autotuning surpasses—and that ceiling remains below the *unoptimised* recognition baseline on both models.

The qualitative character of the autotuned edits reinforces the mechanistic interpretation. The winning base prompt edits were scenario-specific heuristics: prerequisite targeting ("prefer the nearest prerequisite, not the same stuck lecture"), agency preservation ("frame review as a brief reset that returns them to the current lecture"), and collaborative micro-planning ("propose a short next step plus return point"). These are precisely the kind of concrete, prescriptive rules that the active control finding (Section 6.21 of the companion study) showed help weaker models. Recognition, by contrast, operates through general calibration—raising quality across all scenarios without per-scenario tuning.

This distinction—scenario-specific heuristics vs. scenario-general calibration—connects directly to the mechanism model. Autotuning optimises the prompt's *output rules* (what to suggest and how to frame it). Recognition optimises the prompt's *orientation toward the learner* (treating them as an autonomous subject). The super-additivity indicates that these are complementary layers: orientation sets the quality floor; heuristics refine the specific response. Neither substitutes for the other, and both contribute independently to the provable discourse framework's ability to drive measurable improvement.

7.8 Dimension-Targeted Optimisation and Cross-Model Transfer

Section 7.7 demonstrated that automated prompt tuning, guided by the per-dimension rubric, can substantially improve tutoring quality—with recognition prompts more improvable than base prompts, and the gap widening under optimisation. A follow-up experiment extends this finding in three directions: (1) targeting specific rubric dimensions rather than overall score, (2) testing whether modulation-oriented prompt edits can unlock the adaptive responsiveness that Section 6.3 found absent as a general mechanism, and (3) testing whether model-specific prompt optimisations transfer across capability levels.

Interpretive caveat on dimension targeting. The PCA analysis reported in §8.6 shows that the 8 v2.2 tutor dimensions largely measure a single underlying construct (PC1 = 80.7% of variance, Kaiser criterion yields one factor with eigenvalue $>$1, mean inter-dimension r = 0.776). This means "dimension-targeted optimisation" in this section does not isolate empirically independent pedagogical skills; rather, it uses individual rubric dimensions as *interpretive proxies* for aspects of the underlying tutor-quality factor — targeting `adaptive_responsiveness` as a score is operationally a request for the autotuning loop to move the overall quality factor along prompt changes that foreground turn-over-turn adaptation. Results below are correspondingly interpreted as shifts in the single construct via targeted prompt design, not as moving one pedagogical skill while others are held constant. A forced two-factor rotation separates `content_accuracy` from the seven pedagogical dimensions (§8.6), so targeting content accuracy vs. pedagogical quality *is* a meaningful bifactor split; targeting pairs of pedagogical dimensions against each other is not.

7.8.1 Dimension-Targeted Optimisation We extended the autotuning loop with a `--target-dims` parameter that optimises for specific rubric dimensions rather than overall score. Targeting `adaptive_responsiveness` (tutor modulation, 15% weight) and `conceptual_progression` (learner learning, 20% weight) on a single challenging scenario (`mood_frustration_to_breakthrough`, cell 8 bilateral architecture), we ran parallel optimisation sessions on two models: Qwen 3.5 9B (local, via LM Studio) and Claude Haiku 4.5 (OpenRouter). Each session used a hill-climbing loop with an LLM recommender (Claude Opus 4.6) proposing edits, N=1 replication per iteration, and accept/revert decisions based on the composite target dimension score.

Dimension-targeted autotuning results (frustration scenario, cell 8)

	Qwen 3.5 9B	Haiku 4.5
Baseline target dims	35.0	90.0
Best target dims	**75.0** (+114%)	**95.0** (+5.6%)
Adaptive responsiveness	1.75 → 3.75	4.50 → 4.75
Iterations to beat baseline	1	4 (after guidance change)
Lines of prompt added	~300	~60
Key edit	Scaffolded elicitation algorithm	State-change tracking heuristic

The two models required qualitatively different prompt interventions. Qwen's baseline tutor responded to frustrated learners with generic elicitation ("What specifically feels nonsensical?")—a failure mode the 9B model could not resolve without explicit instruction. The winning edit added a highest-priority "Scaffolded Elicitation Rule" mandating 2–3 specific diagnostic probes with substantive content, a superego intervention strategy that flags emotional support without scaffolding as a critical failure, and bilateral learner prompt changes teaching the learner to engage with offered scaffolding. The total addition was ~300 lines across five prompt files, including worked examples and anti-patterns.

Haiku's baseline was already scoring 90 on the target dimensions. Three automated optimisation passes *degraded* performance (90 → 73–85) by adding heavy-handed prescriptive rules—the same over-engineering pattern that helped Qwen actively harmed Haiku by making its output formulaic. The winning pass, which beat the baseline only after operator-provided guidance steering ("be dramatic; try different roles; really provoke the learner"), added just ~60 lines: a single identity statement about tracking state changes, a 6-line adaptive tracking heuristic, and one worked example. The distinction—prescriptive rules vs. tonal permission—maps onto a *prompt-level* version of the prosthesis-straitjacket inversion: weak models need explicit scaffolding rules written into the ego prompt; strong models need tonal permission to exercise capabilities they already possess. This prompt-level pattern is distinct from the *architectural* "cognitive prosthesis" cells 66–68 variant (superego-routed bidirectional profiling), which §6.6.10 tests across six ego models and finds does not follow a capability-threshold curve. The prompt-level asymmetry reported here (Qwen vs Haiku) is compatible with §6.6.10: it is scaffolding in the ego prompt, not superego-routed profiling, and it carries no implication that architectural bidirectional profiling would help a weak model.

Both sessions converged on a **bilateral fix pattern**: tutor and learner prompts changed together. Qwen's fix taught the tutor to scaffold and the learner to engage with scaffolding. Haiku's fix taught the tutor to track state changes and the learner to show cumulative progression. Unilateral changes (tutor-only or learner-only) were less effective in both cases.

7.8.2 Can Prompt Optimisation Create Adaptive Responsiveness? The M3 null finding (Section 6.3: all slope $d \leq 0.15$) was established under unoptimised prompts. The dimension-targeted experiment provides evidence that adaptive responsiveness can be *created through prompt optimisation* on weak models and *enhanced* on strong ones—but the mechanism differs.

Qwen's best transcript (iteration 6) shows a genuine four-stage modulation arc: scaffolding with diagnostic options (Turn 0) → tracking the learner's specific choice (Turn 1) → rescaffolding with sub-anchors within the chosen topic (Turn 2) → stepping back to respect learner autonomy (Turn 3). This strategy shift across turns was absent at baseline and appeared only after explicit scaffolding rules were added to the prompt. The prompt *created* modulation by providing an algorithm the model could follow.

Haiku's best transcript shows a different kind of modulation: direct challenge (Turn 0) → naming a specific state change (Turn 1) → assigning a concrete task tied to the learner's own question (Turn 2) → *reframing what breakthrough means* (Turn 3). Turn 3 is particularly revealing: the tutor adapts its *interpretation* of the learner's state, not just its tracking of it. This modulation was already latent in Haiku's capabilities; the prompt merely encouraged it.

This result does not contradict the M3 null finding. The pooled analysis (N=432) tested whether *experimental conditions* (recognition vs. base, single vs. multi-agent) produce differential trajectory slopes. They do not. But within a single condition, individual prompt optimisation can create or enhance turn-over-turn adaptation—particularly on scenarios with clear emotional arcs and sustained multi-turn demands. This prompt-lab finding is itself small-N and model-specific; it is separate from the M3 mechanism question, which is decisively null on both the primary analysis and on A12's pre-registered replication of the exploratory DeepSeek/Sonnet disengagement effect (§6.3.2).

7.8.3 Cross-Model Prompt Transfer If prompt optimisation is model-specific (as Section 7.8.1 implies), then optimised prompts should not transfer well across capability levels. We tested this directly by running each model's best-performing prompts on the other model (N=3 replications each, same scenario and cell, judged by Sonnet 4.6).

Cross-model transfer matrix (target dimension scores, frustration scenario)

	Qwen-optimised prompts	Haiku-optimised prompts	Baseline prompts
Qwen 3.5 9B	**75.0** (native)	**52.2** (transfer)	35.0
Haiku 4.5	**62.2** (transfer)	**95.0** (native)	90.0

The transfer results are asymmetrically destructive. Heavy-to-strong transfer (Qwen-optimised prompts on Haiku) produces a **27.8-point regression** below Haiku's unoptimised baseline (62.2 vs. 90.0)—the prescriptive scaffolding rules that served as a cognitive prosthesis for Qwen become a cognitive straitjacket for Haiku, overriding its natural ability to read the situation and respond adaptively. Light-to-weak transfer (Haiku-optimised prompts on Qwen) produces a **17.2-point improvement** over Qwen's baseline (52.2 vs. 35.0) but falls **22.8 points short** of Qwen's native optimisation (52.2 vs. 75.0)—the lightweight heuristics help somewhat but provide insufficient scaffolding for the weak model.

The asymmetry is theoretically informative. Permission-based edits (Haiku-style) are **safe but insufficient** for weak models: they do not actively harm, but they leave capability on the table. Rule-based edits (Qwen-style) are **high-reward on target but destructive off-target**: they unlock capability on weak models but degrade strong ones. This suggests an inverted-U relationship between prompt complexity and model capability—not merely diminishing returns, but active harm beyond the optimum. The practical implication is that prompt optimisation must be model-stratified: a universal "best prompt" does not exist across capability levels, and deploying heavy scaffolding prompts on models that do not need them risks worse-than-baseline performance.

This finding connects to the prompt density alternative explanation (Section 7.9): if prompt elaboration were uniformly beneficial, the Qwen-optimised prompts (which add ~300 lines of detailed instruction) should improve Haiku. Instead, they degrade it catastrophically. The result provides direct experimental evidence that prompt *content and targeting* matter more than prompt *volume*—and that the relationship between prompt complexity and output quality is non-monotonic.

7.8.4 Implications for Superego Redundancy The universal substitution finding (Section 6.4: superego benefit collapses to near-zero under recognition on strong models) is reinforced and sharpened by the dimension-targeted optimisation experiment. The Qwen session's winning edit included a new superego intervention strategy ("Scaffolding Intervention") that catches a specific failure mode—emotional validation without conceptual scaffolding—and forces ego revision. This designed-in error correction was part of the bilateral fix that lifted Qwen's adaptive responsiveness from 1.75 to 3.75. By contrast, the Haiku session's failed passes (1–3) added analogous superego strategies, and every addition *degraded* performance: the strong model's already-calibrated output did not benefit from additional correction, and the superego rules made the output formulaic rather than adaptive.

This sharpens the substitution claim in both directions. On weak models, the superego is not merely "somewhat useful" (the Gemini Flash 3.0 residual of +12.3, Section 6.4.1)—its value can be substantially amplified by designing targeted intervention strategies for specific failure modes. The baseline superego provides generic error correction; a prompt-optimised superego provides *failure-mode-specific* correction that the weak model's ego cannot achieve through calibration alone. On strong models, the superego is not merely "redundant" (near-zero delta)—actively adding superego scaffolding is *destructive*, degrading performance below baseline. The gradient from helpful to harmful runs through the same capability threshold that governs the prosthesis-straitjacket inversion.

One limitation: the Qwen superego fix worked as part of a bilateral change (ego, superego, and learner prompts all changed together), so the superego's independent contribution cannot be isolated from this experiment. The factorial design (Section 6.4) provides the controlled isolation; the prompt lab provides a worked example of how to amplify the superego's value when it does have a role.

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